

Dieter Banaag

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EXPERIENCE

Software Engineering Intern

Jan. 2025 – Aug. 2025

Procter & Gamble

Toronto, ON

- Designed and implemented automated workflows for employee onboarding and offboarding using Java, Python, and Selenium, reducing manual processing time from 20 to 3 minutes per employee.
- Automated form and ticket processing pipeline, reducing average resolution time by over 80% while maintaining full audit compliance.
- Built a Power BI sales dashboard to track 20+ metrics across 3 regions, enabling faster reporting for stakeholders.
- Automated PDF-to-Excel conversion with proof-of-performance reconciliation across 10,000+ products per invoice, eliminating 15 hours of manual work per month.
- Increased unit test coverage from 5% to 75% using Java/Selenium, reducing regression bugs in production.

Software Developer Intern

May 2024 – Dec. 2024

Mercedes and Singh(Architectural design studio & SaaS product lab)

Calgary, AB

- Built site selection and 3D visualization tools for Home.ai, a SaaS residential design platform, using React, ArcGIS, and Three.js.
- Integrated GeoJSON datasets for parcels and zoning, enabling accurate design decisions.
- Implemented Firebase authentication and secure database design.
- Automated permit-ready file generation, reducing design timelines by 6x.

PROJECTS

Itinerary Planning Platform – Full Stack Developer

2026 – Present

Full-stack travel planning app with real-time collaboration and external API integrations.

- Built end to end with a React frontend and Node/Express backend, owning the UI, API layer, and data model.
- Designed modular CRUD flows for trips, days, activities, budgets, and reviews using reusable contexts, controllers, and services.
- Implemented Firebase Auth with server-side token verification and member-based access control for collaborative planning.
- Integrated geocoding, weather, prayer times, and place search APIs through a dedicated backend enrichment layer.
- Added real-time Firestore subscriptions and optimistic updates to keep itinerary state and chat responsive across users.

Teacher-First Classroom Management Platform (Full Stack)

Dec. 2025 – Present

- Built a full-stack classroom management app with Next.js, TypeScript, and React, backed by a PostgreSQL (Supabase) schema with role-based access, ownership constraints, and referential integrity.
- Implemented secure auth via Firebase Auth with server-side token verification, protected API routes, and clean architecture separating UI, business logic, and backend services.

VR Operational Preparedness Simulator (Hardware & Software Engineer)

Sept. 2025 – Apr. 2026

- Developing a mixed-reality training simulator for oil & gas operations, combining VR with a custom-built physical control panel (circuit design, soldering, and wiring) for immersive, tactile learning.
- Developed and flashed firmware on ESP32 microcontrollers to bridge physical sensor/control signals to VR application state in real time.
- Awarded Bronze Medal out of 26 competing teams at the Multidisciplinary Capstone Competition

EDUCATION

University of Calgary

Sep. 2021 – Apr. 2026

B.S. Software Engineering · Calgary, AB

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks & UI: React, Next.js, Node.js, Tailwind, Shadcn/ui, Material-UI, Sass, JUnit

Cloud & Tools: AWS, Docker, Git, Firebase, Power BI, VS Code, UNIX

Databases: MySQL, PostgreSQL (Supabase), MongoDB

Libraries: pandas, NumPy, scikit-learn

Other: ArcGIS, Three.js, WebAssembly, UML/EER Modeling